

Users' Awareness and Perceptions of Smartphone Privacy and Security Settings: A Replication Study in Saudi Arabia

Sara Abdulrahman Bajarsh
Saudi Arabia
saraayb2003@gmail.com

Khuluod Theyab Alharbi
Saudi Arabia
khuluodalharbi@gmail.com

Lama Ahmed Alghamdi
Saudi Arabia
lama.asg0@gmail.com

Dr. Elham Al Qahtani
Saudi Arabia
ealqahatni@uj.edu.sa

Abstract

Smartphone privacy and security settings are key tools for helping users manage digital risks, yet most prior work has examined these settings in Western contexts. We replicate and extend Frik et al. [30] in the Saudi context to examine how Saudi smartphone users understand, configure, and evaluate privacy and security settings. Through an online survey with 119 Saudi smartphone users, we studied awareness, default beliefs, past configuration, future intentions, perceived difficulty, perceived effectiveness, usability impact, and risk concerns across 18 smartphone settings. We find that participants were generally aware of common settings, but this awareness did not always translate into accurate understanding or prior configuration. Less visible settings, such as erase data, storing audio, and password reuse detection, showed lower awareness and use, although participants reported willingness to configure them in the future. Participants also perceived most settings as effective, but expressed stronger concerns about hacking, unauthorized access, phishing, scams, personal data disclosure, and location tracking. Our findings confirm the awareness-behavior gap identified in prior work while showing that privacy and security perceptions in Saudi Arabia are shaped by cultural expectations, institutional trust, and Arabic privacy terminology. We discuss implications for culturally localized privacy interfaces and Saudi-focused awareness efforts for Arabic-speaking users.

Keywords

smartphone privacy, smartphone security settings, Saudi Arabia, usable privacy and security, privacy awareness, replication study

1 Introduction

Smartphones have become essential tools for communication, education, healthcare, banking, social networking, and access to digital services. These devices store sensitive information such as location data, biometric identifiers, financial records, contacts, and personal communications. Therefore, smartphone privacy and security settings—including app permissions, biometric authentication, location controls, tracking restrictions, browser privacy controls, and security notifications—play an important role in protecting users'

personal information [23, 27, 30, 38, 44, 45, 47, 59]. However, previous research has shown that users often do not fully understand these settings, may not know where to find them, and frequently fail to configure them effectively [23, 27, 30, 38, 59].

This issue is particularly relevant in Saudi Arabia, where internet penetration has reached approximately 99%, and mobile devices serve as the primary means of accessing digital services [16]. Saudi Vision 2030 has accelerated digital transformation across government, healthcare, education, finance, and identity services [21?]. Official digital platforms such as Absher, Tawakkalna, Nafath, and Sehhaty increasingly rely on smartphones for identity verification, location-based services, health information management, and access to public services [21, 62]. Consequently, Saudi smartphone users routinely make privacy and security decisions that may affect the protection of their personal data.

In parallel with this digital transformation, cybersecurity awareness has become a national priority. Previous Saudi studies suggest that awareness of cybersecurity risks does not always translate into consistent protective behavior, highlighting the need to examine the relationship between awareness, self-efficacy, and actual privacy-setting configuration among Saudi smartphone users [11, 13].

Although Frik et al. [30] investigated users' understanding and use of smartphone privacy and security settings, their study was conducted in a Western context. This creates a cultural and contextual gap because privacy in Saudi society may be influenced by family expectations, religion, social reputation, gender norms, and collective cultural values [3, 5, 25]. Saudi users may therefore evaluate location sharing, biometric authentication, photo access, camera permissions, tracking, and personal data disclosure differently from users in Western societies. Furthermore, many Saudi users may rely on family members or trusted friends when making technology-related decisions, making social influence an important factor in privacy behavior [3, 10].

Replicating the study in Saudi Arabia is justified for three reasons. First, there is a cultural gap, as privacy expectations in Saudi Arabia often extend beyond individual concerns to include family privacy, social visibility, and reputation [3, 5, 25]. Second, rapid digital transformation has increased dependence on smartphone-based services, digital identity systems, biometric authentication, and location-enabled applications [16, 21, 62?]. Third, language and interface design may affect user understanding, as Arabic translations of technical concepts such as tracking, permissions, and two-factor authentication may influence users' confidence and ability to configure settings correctly [10, 30, 57].

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Similar to the original study, this research also considers information architecture, as privacy and security settings are often distributed across multiple menus in iOS and Android operating systems. Such fragmentation may make it difficult for users to locate, understand, and configure these settings, particularly when technical terminology is translated into Arabic or presented without sufficient explanation [10, 30, 57, 59, 72].

Following the design of Frik et al. [30], this study replicates and adapts their survey to the Saudi cultural and technological context. Specifically, the study addresses the following research questions:

- RQ1: To what extent are Saudi smartphone users aware of the privacy and security settings available on their smartphones and their default configurations?
- RQ2: To what extent have Saudi smartphone users configured smartphone privacy and security settings in the past?
- RQ3: To what extent do Saudi smartphone users intend to configure smartphone privacy and security settings in the future?
- RQ4: To what extent do Saudi smartphone users perceive themselves capable of configuring smartphone privacy and security settings, and what difficulties do they anticipate?
- RQ5: How do Saudi smartphone users perceive the impact of privacy and security settings on smartphone usability and their effectiveness in mitigating privacy and security risks?
- RQ6: How do Saudi cultural and demographic characteristics—including age, gender, education, digital experience, family influence, social norms, and privacy expectations—affect awareness, behaviors, intentions, self-efficacy, and perceptions regarding smartphone privacy and security settings?

To answer these questions, we conducted a survey of 119 Saudi smartphone users from diverse demographic backgrounds. The survey was adapted from the original study and covered 18 smartphone privacy and security settings, including location services, passcodes, biometric authentication, notification previews, automatic updates, ad personalization, two-factor authentication, analytics, cookies, browser tracking, and other privacy-related controls. The contact tracing setting included in the original study was excluded because it was specific to the COVID-19 period and is no longer commonly relevant to current smartphone privacy practices.

The findings indicate that participants were generally aware of common privacy and security settings; however, their understanding of default configurations and confidence in changing settings varied considerably. The results further suggest that cultural factors, language clarity, family influence, and digital experience may affect how Saudi users understand and manage smartphone privacy and security settings. Based on these findings, this study provides recommendations for improving Arabic privacy terminology, simplifying smartphone privacy interfaces, and designing cybersecurity awareness programs that better reflect the Saudi context.

2 Original Study Overview

Frik et al. [30] investigated how smartphone users understand, use, and make decisions regarding privacy and security settings. The study was motivated by the observation that smartphones provide numerous built-in protections, including location controls, passcodes, biometric authentication, two-factor authentication, automatic updates, browser privacy settings, and tracking controls.

However, these protections are only effective when users are aware of them, understand their default configurations, can locate them within settings menus, and feel confident enough to configure them appropriately [30].

To examine these issues, Frik et al. conducted an online survey involving 178 smartphone users in the United States from diverse demographic backgrounds. The survey covered 19 common privacy and security settings available on iOS and Android devices. Participants were asked about their awareness of these settings, their understanding of default configurations, whether they had configured them previously, whether they intended to configure them in the future, the anticipated difficulty of changing them, and their perceptions of the effectiveness of these settings in mitigating privacy and security risks. The study also examined differences across demographic characteristics, including age, gender, education, income, race/ethnicity, and digital experience [30].

The findings revealed that many users were unaware of several available privacy and security settings and frequently misunderstood their default configurations. Although participants expressed concerns about privacy and security risks, many had not configured available protections because they expected difficulties in locating or changing settings, lacked technical knowledge, or believed that modifications could negatively affect usability. The study further identified demographic differences, suggesting that certain groups—including older adults, women, and racial or ethnic minorities—faced greater challenges in engaging with privacy and security settings and anticipated greater difficulty in configuring them. Based on these findings, Frik et al. recommended improvements in the organization, clarity, and usability of smartphone privacy and security settings to support more informed user decision-making [30].

3 Related Work

Smartphones have become one of the most widely used internet-connected technologies in Saudi Arabia. Consequently, privacy and security settings represent one of the primary mechanisms through which users express their privacy and security preferences and protect their personal information. Therefore, the usability of these settings is critical for ensuring effective privacy and security protection. Similar to Frik et al. [30], we first review existing research on the usability of smartphone privacy and security settings. We then discuss prior findings regarding privacy and security perceptions, behaviors, and experiences across different socio-demographic groups, with particular attention to evidence from Saudi Arabia and the Arab region.

3.1 Usability of Privacy and Security Settings

While research on mobile application permissions has made substantial progress, the usability of smartphone privacy and security settings remains comparatively underexplored. For example, Ramakapane et al. [59] examined smartphone manufacturer-provided privacy features and found that many users were unaware of default privacy configurations and often accepted default settings without understanding their implications. Similarly, Egelman et al. [22] and Harbach et al. [32] reported that many users perceive screen-locking mechanisms as inconvenient, which discourages

the adoption of available security protections. Micallef et al. [51] further demonstrated that context-aware authentication mechanisms can improve the perceived usability of smartphone locking by reducing unnecessary authentication requirements.

However, these studies primarily focused on individual security mechanisms such as authentication or default privacy features rather than the broader ecosystem of smartphone privacy and security settings. Moreover, they largely examined the convenience of using specific protections rather than users' awareness of available settings, beliefs about default configurations, perceived self-efficacy, and willingness to configure such settings.

Zhou et al. [72] developed and evaluated a prototype application that guided users through smartphone privacy configurations and found that users generally desired greater control over privacy settings than currently provided by mobile operating systems. The study also reported that older participants were less likely to actively seek information about privacy protections. Nevertheless, the research relied on a prototype environment and focused on a limited set of privacy features rather than actual smartphone settings. Similarly, Vecchiato et al. [68] proposed an automated assessment tool for Android security configurations and observed that users rarely modify factory-default settings. However, their work focused primarily on Android security settings and did not investigate users' perceptions, motivations, or beliefs regarding those configurations.

In the Saudi context, examining the usability of smartphone privacy and security settings is especially important because users' decisions may be shaped not only by interface complexity, but also by Arabic terminology, digital literacy, trust in official platforms, and reliance on family members or peers for technical support [10, 15, 16, 31].

3.2 Socio-Demographic Differences in Privacy and Security

Prior work has consistently demonstrated the existence of a privacy knowledge-behavior gap, whereby individuals express concern about privacy risks but do not necessarily engage in protective behaviors. Crossler and Bélanger [19] found that knowledge about smartphone privacy protections does not automatically translate into the adoption of location-protective settings. Similarly, Ketelaar and Van Balen [39] reported that smartphone literacy plays a significant role in shaping privacy-related attitudes and behaviors, while Park [57] showed that higher levels of digital literacy are associated with stronger online privacy protection practices. These findings suggest that privacy self-efficacy and knowledge are important determinants of protective behavior, yet they may not be equally distributed across user populations.

Research has also shown that demographic characteristics influence privacy and security outcomes. Studies conducted in Western contexts reported differences associated with age, gender, education, and technology literacy [19, 39, 57, 58, 63]. Older adults often report lower awareness of privacy risks and lower confidence in managing security settings, while users with greater technological experience generally exhibit stronger privacy-protective behaviors [57]. Nevertheless, prior studies also indicate that knowledge

alone is insufficient, as even users with moderate or high security knowledge may still engage in risky behaviors [24, 65].

Within the Arab region, privacy and security behaviors are shaped by additional cultural and social considerations. Abokhodair and Vieweg [5] found that privacy management in Gulf societies is closely connected to family relationships, social reputation, and cultural expectations. Subsequent studies conducted in Saudi Arabia, Qatar, and other Gulf countries further demonstrated that users frequently balance technological benefits against concerns related to family visibility, social norms, and personal reputation [2, 3, 10]. Research on youth social media use similarly suggests that privacy concern and protective behavior are shaped by social relationships and platform norms [69]. More recently, Alashwali and Cranor [10] examined privacy perceptions among Saudi users and reported growing awareness of privacy issues, while Farooq et al. [25] identified widespread privacy concerns across Middle Eastern and North African countries. Evidence from the United Arab Emirates also suggests that privacy paradox behaviors observed in Western populations are similarly present among Arab users [17].

Collectively, the existing literature suggests that awareness, self-efficacy, perceived usability, and socio-demographic characteristics all influence privacy and security behavior. However, limited research has examined these factors within the context of smartphone privacy and security settings in Saudi Arabia. Therefore, this study replicates and extends the work of Frik et al. [30] by examining Saudi users' awareness, past configuration behavior, future intentions, perceived self-efficacy, perceived effectiveness, expected usability impact, privacy concerns, and socio-demographic differences regarding smartphone privacy and security settings.

4 Method

This study adopted a descriptive replication approach to explore the gap between users' awareness of smartphone privacy and security settings and their actual configuration behaviors among users in Saudi Arabia, building on the framework introduced by Frik et al. [30]. The study focused on contextual factors specific to the Saudi environment, particularly cultural and linguistic variables surrounding privacy and technology use that have not been sufficiently addressed in prior Western-centered studies. Data were collected through an online survey administered using Google Forms. Participants were recruited through social media platforms among Saudi nationals who use Android or iOS mobile operating systems. At the beginning of the survey, screening questions were used to ensure that all participants were Saudi citizens, aged 18 years or older, and active smartphone users. To achieve a diverse sample of participants, demographic information was collected regarding age group, gender, administrative region, educational level, employment status, and smartphone operating system. Upon accessing the survey, participants were directed to a Google Forms page to complete the questionnaire. We obtained IRB approval from the Deanship of Scientific Research prior to conducting the study. All participants provided informed consent before responding to the survey.

4.1 Settings Selection

To survey participants about common smartphone privacy and security settings, we first generated a list of privacy and security settings available in the current versions of iOS and Android at the time of conducting this study. Similar to Frik et al. [30], we focused on settings provided directly by the mobile operating systems and their default browsers, Safari for iOS and Chrome for Android. Because of the heterogeneity in use and availability of settings across individual applications, app-specific settings were excluded from the analysis to support broader comparison across devices. The default browsers were included because they are pre-installed on users' smartphones and are commonly used for browsing and accessing online services [30, 63].

The initial list of settings was reviewed based on privacy and security relevance and updated to reflect changes in current iOS and Android functionality. Following prior work on usable privacy and security settings [8, 18, 28, 30, 71], and research on app permissions and user decision-making [26, 37], the selection focused on settings related to user control over personal data, device access, browsing activity, account protection, app permissions, voice assistant privacy, and file sharing. COVID-19 contact tracing settings were excluded because they are no longer consistently available or actively used in current smartphone operating systems.

The final list included 18 groups of privacy and security settings (Table 1). Compared with the original study, the list was updated to reflect current smartphone settings while preserving comparability with Frik et al. [30]. This approach ensured that the selected settings remained relevant to the current smartphone environment and suitable for examining privacy and security expectations and behaviors among smartphone users in Saudi Arabia.

4.2 Survey

In the first part of the survey, we asked participants an open-ended question about two privacy or security risks they worry about when using their smartphones; participants were encouraged to consider different actors who may introduce those risks, including companies, government entities, someone they know, or strangers. We then asked them to describe, without looking at their phones, any settings currently available on their smartphones that they could configure to address each of the privacy or security risks they mentioned. Participants were instructed to write "N/A" if they believed no such settings were available. The goal of this part was to elicit unaided recall of privacy and security risks and related smartphone settings.

In the second part of the survey, participants were asked a series of questions about the 18 smartphone privacy and security settings included in this study. For each setting, we asked whether participants believed the setting was an existing option on their smartphone, what they believed its default configuration to be, how easy or difficult it would be to configure, and reasons for any expected difficulty. We also asked whether they had ever configured the setting before, their intentions to configure it in the future, what privacy or security risks they believed the setting would protect them from, how concerned they were about those risks, the perceived effectiveness of the setting, and the expected impact of configuring the setting on the overall smartphone experience.

The survey instrument was adapted from Frik et al. [30]. Prior to data collection, all survey questions and response options were translated from English into Arabic and reviewed by the research team to ensure linguistic clarity and preserve conceptual consistency with the original instrument in the Saudi context. Finally, participants were asked about demographic characteristics, including age, gender, education, income, employment status, and region of residence, in addition to digital skills, technical background, smartphone usage habits, and the model and operating system of their smartphones. The complete survey instrument can be found in Appendix A.

4.3 Analysis

For the quantitative analysis, we used descriptive statistics and regression models to examine participants' responses across the 18 privacy and security settings. Logistic regression was used for categorical outcomes, while ordered logistic regression was used for ordinal Likert-scale responses, including perceived difficulty, future intentions, perceived effectiveness, risk concern, and expected impact on user experience. The analysis considered participants' awareness of settings, beliefs about default configurations, prior configuration behaviors, future intentions, perceived difficulty, perceived effectiveness, and privacy/security concerns, along with demographic and background variables such as age, gender, region, education, income, operating system, smartphone use, and technical background.

For the qualitative analysis, we analyzed open-ended responses using thematic coding. Since the survey was administered in Arabic, participants' responses were first reviewed in Arabic to preserve meaning, then translated into English where needed for coding, reporting, and comparison with Frik et al. [30]. Two researchers independently reviewed the responses and developed initial codebooks for the open-ended questions. They then discussed and merged the codes into a unified coding scheme, resolving disagreements through discussion. The final codebooks were applied to all responses to identify recurring themes related to privacy and security risks, participants' understanding of available settings, and reasons for perceived configuration difficulty. This approach follows the qualitative analysis procedure used in prior usable privacy and security research [30, 32].

4.4 Limitations

As with any survey-based study, this research has several limitations. First, the study relied on self-reported responses, which may be influenced by recall bias or overestimation of participants' awareness and familiarity with smartphone privacy and security settings. Because the survey aimed to examine users' perceptions, expectations, and prior experiences, participants responded based on recall rather than by checking their devices directly. While this approach is consistent with prior work examining users' mental models and privacy expectations [6, 30], reported responses may not fully reflect actual device configurations or observed behavior. Second, although the survey included participants from different demographic groups and administrative regions across Saudi Arabia, the sample size remains limited and does not represent all smartphone users in the country. In addition, the 18 selected settings were

Table 1: Privacy and security settings used in the survey. Asterisk (*) indicates settings with platform-specific wording or interface differences across operating systems.

Setting	Description shown to iOS users	Description shown to Android users
Location services		Turn off location services
Passcode	Create a passcode to unlock the smartphone and authorize actions like payments or password autofill	
Face unlock	Only allow the use of face recognition to unlock the smartphone if you are looking at the screen	
Notification preview	Allow the smartphone to show a preview of notification content only when it is unlocked	
Automatic updates	Turn on automatic updates of the smartphone’s operating system	
Erase data*	Turn on erasing of all data on the smartphone after 10 failed passcode attempts	
Ad personalization	Limit ad tracking and ad personalization on your smartphone	
Two-factor authentication	Enable two-factor authentication for your iCloud account	Enable two-factor authentication for your Google account
Analytics	Deny analytics of usage and data on your iCloud account	Deny analytics of usage and data on your Google account
Voice commands	Don’t allow the smartphone to listen for “Hey Siri” commands	Don’t allow the smartphone to listen for “Hey Google” commands
Storing audio	Don’t allow storing audio recordings of your Siri and Dictation interactions	Don’t allow storing audio recordings of your Google Assistant interactions
Password reuse	View passwords that have been re-used on other websites in your smartphone’s Safari browser	View passwords that have been re-used on other websites in your smartphone’s Chrome browser
Pop-ups	Block pop-up windows on websites in your smartphone’s Safari browser	Block pop-up windows on websites in your smartphone’s Chrome browser
Website warnings*	Turn on fraudulent website warnings in the smartphone’s Safari browser	Turn on fraudulent website warnings in the smartphone’s Chrome browser
Cookies	Block all cookies in the smartphone’s Safari browser	Block all cookies in the smartphone’s Chrome browser
Browser tracking	Limit tracking of your activity in the smartphone’s Safari browser	Limit tracking of your activity in the smartphone’s Chrome browser
Voice assistant ad personalization*		Limit ad tracking and ad personalization in the smartphone’s voice assistant
AirDrop / file sharing*	Allow only your contacts to share files with you through AirDrop or built-in file sharing	Allow only your contacts to share files with you through built-in file sharing

distributed across six survey versions, with each participant answering questions about three settings only. This design reduced survey fatigue and supported survey completion; however, responses were therefore distributed across settings rather than collected from all participants for every item. Accordingly, the findings should be interpreted within the scope of the study sample rather than as population-level estimates [30, 60]. Third, while the study covered commonly used smartphone privacy and security settings and accounted for recent operating-system updates at the time of data collection, the selected settings do not capture the full range of privacy and security controls available across devices and applications. Future research may extend this work by examining additional settings and app-specific privacy controls and by further exploring smartphone privacy and security behaviors in broader real-world contexts within Saudi Arabia and the Arab region [10, 25].

5 Results

We collected a total of 119 valid responses through an online survey distributed via social media platforms. Responses were screened according to the study’s inclusion criteria, resulting in a final sample of Saudi smartphone users. The analysis includes 18 smartphone

privacy and security settings. To support comparison with Frik et al. [30], we analyzed participants’ awareness, prior configuration behavior, future intentions, perceived difficulty, perceived effectiveness, and expected usability impact across the selected settings.

5.1 Participants

The final sample consisted of 119 Saudi smartphone users. Participants represented a range of age groups, with 35.3% aged 18–24 years, 31.9% aged 25–34 years, 16.8% aged 35–44 years, 11.8% aged 45–54 years, and 4.2% aged 55 years or older. Participants were recruited from all major regions of Saudi Arabia, including the Western (39.5%), Central (16.8%), Eastern (16.8%), Northern (14.3%), and Southern (12.6%) regions.

Table 2 summarizes participants’ age, region of residence, education, employment, and household income characteristics.

Table 2: Participant characteristics based on survey responses ($N = 119$).

Individual characteristics	N	%
<i>Age group</i>		
18–24 years	42	35.3%
25–34 years	38	31.9%
35–44 years	20	16.8%
45–54 years	14	11.8%
55+ years	5	4.2%
<i>Region of residence</i>		
Central region	20	16.8%
Western region	47	39.5%
Eastern region	20	16.8%
Northern region	17	14.3%
Southern region	15	12.6%
<i>Education level</i>		
High school or below	13	10.9%
Diploma	5	4.2%
University student	28	23.5%
Bachelor’s degree	55	46.2%
Master’s degree	15	12.6%
Doctoral degree	3	2.5%
<i>Employment</i>		
Full-time employee	50	42.0%
Student	34	28.6%
Part-time employee	11	9.2%
Unemployed and looking for work	7	5.9%
Unemployed and not looking for work	6	5.0%
Retired	3	2.5%
Other	8	6.7%
<i>Household income</i>		
Less than 5,000 SAR	2	1.7%
5,000–<10,000 SAR	15	12.6%
10,000–<15,000 SAR	19	16.0%
15,000–<20,000 SAR	24	20.2%
20,000–<30,000 SAR	19	16.0%
30,000+ SAR	23	19.3%
Prefer not to answer	17	14.3%

Regarding education, most participants were either university students (23.5%) or held a bachelor’s degree (46.2%), while smaller proportions reported diploma-, master’s-, doctoral-, or high-school-level education. In terms of employment, 42.0% were employed full-time and 28.6% were students, with the remaining participants distributed across part-time employment, unemployment, retirement, and other categories.

Similar to Frik et al. [30], our goal was to recruit participants with diverse demographic backgrounds rather than to obtain a nationally representative sample of smartphone users. The resulting sample therefore provides variation in age, education, employment status, income, and geographic location, supporting the study’s examination of differences in smartphone privacy and security awareness, behaviors, and perceptions across demographic groups.

5.2 Free-text Responses about Privacy and Security Risks and Settings

We first asked participants about the smartphone privacy and security risks that worried them (Q3) and what settings they believed could help mitigate those risks (Q4). These open-ended questions were presented before the structured multiple-choice questions to capture participants’ unaided recall and avoid priming them with predefined categories of risks or settings.

Risks. Participants most frequently expressed concerns about hacking, account compromise, and unauthorized access to personal data. Many responses referred to stolen passwords, unauthorized access to photos, and account takeovers, especially for banking, government, or social media accounts. This reflects the importance of smartphones in Saudi Arabia as access points to digital identity, financial services, and government platforms.

Another common concern involved data collection, leakage, and misuse by applications or companies. Participants worried that apps may collect personal information without clear consent, share data with third parties, or use it for tracking and advertising. Location tracking was also frequently mentioned, with participants expressing concern about being monitored or having their movements accessed without permission. These concerns are especially relevant in the Saudi context, where location-based services and mobile public-service applications are widely used.

Participants also mentioned phishing, suspicious links, online shopping scams, and financial fraud, including unauthorized transactions or access to banking information. Some responses were general or vague, such as concerns about “privacy violation” or “data exposure,” while others indicated limited awareness of specific privacy or security risks.

Settings. When asked what smartphone settings could help mitigate these risks, participants most often mentioned authentication-related protections, including two-factor authentication, strong passwords, passcodes, and biometric authentication. These settings were mainly associated with protecting accounts and preventing unauthorized access.

Participants also referred to privacy controls that limit data collection, such as restricting app permissions, disabling location access, turning off tracking, and limiting access to photos, contacts, microphones, or cameras. Some participants suggested broader security practices, such as avoiding suspicious links, not using public Wi-Fi networks, and being careful when downloading apps. Similar to Frik et al. [30], these answers reflect that some users think about protective behavior more than specific configurable smartphone settings.

However, a noticeable portion of participants could not identify any specific settings that would address their concerns, often responding with “none,” “I do not know,” or vague answers. This suggests a gap between users’ awareness of privacy and security risks and their knowledge of the smartphone settings that could help mitigate those risks.

5.3 Awareness about the Settings

To answer RQ1, we asked participants to indicate, without looking at their phones, whether they believed each privacy or security setting was available on their smartphone (Q5). Overall, participants

showed high awareness of several common settings, but awareness varied across the 18 settings (Figure 1).

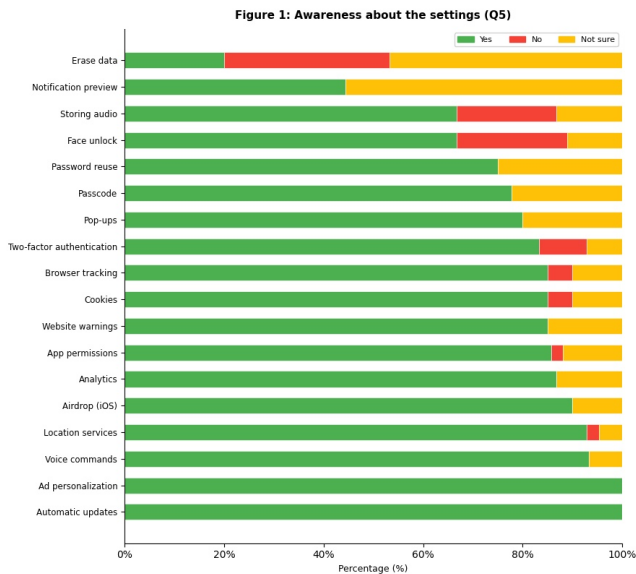


Figure 1: Participants' awareness of smartphone privacy and security settings (Q5).

Participants were most aware of automatic updates and ad personalization, where all respondents indicated that these settings existed on their smartphones. Awareness was also high for voice commands, location services, AirDrop, analytics, app permissions, website warnings, cookies, browser tracking, and two-factor authentication. These findings suggest that Saudi smartphone users are generally familiar with settings that are frequently encountered in daily smartphone use or commonly discussed in privacy and security contexts.

However, awareness was lower for less visible or more technical settings. Erase data had the lowest awareness, with only about one-fifth of participants indicating that the setting existed, while many selected "No" or "Not sure." Notification preview also showed high uncertainty, and participants were less certain about storing audio and face unlock compared with more familiar controls.

Following the structure of Frik et al. [30], these results show that awareness is uneven across smartphone privacy and security settings. In the Saudi context, this uneven awareness may be influenced by how settings are translated, labeled, and surfaced in Arabic interfaces. Settings that are hidden deeper in the system or described using technical terminology may be less recognizable, even when they are relevant to users' everyday privacy concerns.

5.4 Beliefs about Default Settings

To answer RQ1, we asked participants what they believed the default configuration of each setting would be if they had just bought a new smartphone and had not changed any settings yet (Q6). Overall, participants' beliefs about default settings varied across the 18 settings (Figure 2).

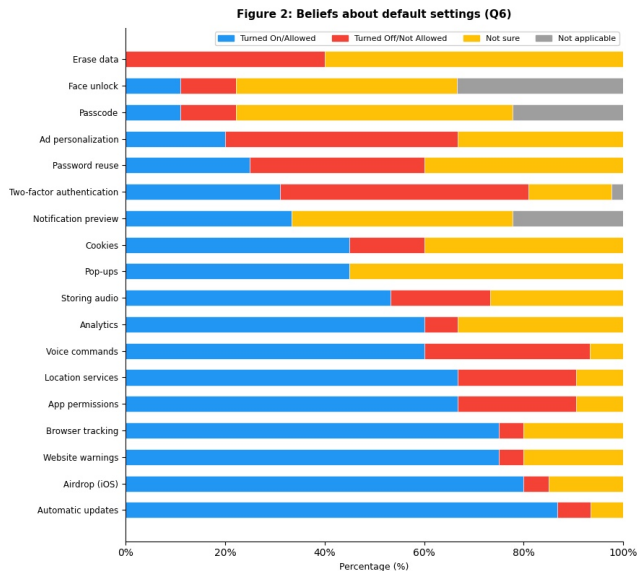


Figure 2: Participants' beliefs about default configurations of smartphone privacy and security settings (Q6).

Participants were most confident that automatic updates, AirDrop, website warnings, browser tracking, app permissions, and location services were turned on or allowed by default. In contrast, uncertainty was high for erase data, passcode, face unlock, notification preview, pop-ups, and password reuse. For erase data, most participants were either unsure or believed it was turned off by default, suggesting limited familiarity with this security feature.

Participants also showed uncertainty about authentication-related settings such as passcodes, face unlock, password reuse alerts, and two-factor authentication. This is important because these settings are closely related to account and device protection, yet users may not clearly understand whether they need to activate or configure them manually.

These findings indicate that awareness of a setting does not necessarily mean understanding its default configuration. In the Saudi context, this uncertainty may be shaped by differences between iOS and Android, the complexity of privacy menus, and the clarity of Arabic terminology used to describe default privacy and security behaviors.

5.5 Past Behaviors and Intentions to Configure Settings

To answer RQ2 and RQ3, we asked participants whether they had configured each setting in the past (Q9) and whether they intended to configure it in the future (Q12). Figures 3 and 4 summarize the responses. Overall, participants reported high past engagement with several common settings, while less visible settings showed lower prior configuration. However, future intentions were generally high across most settings.

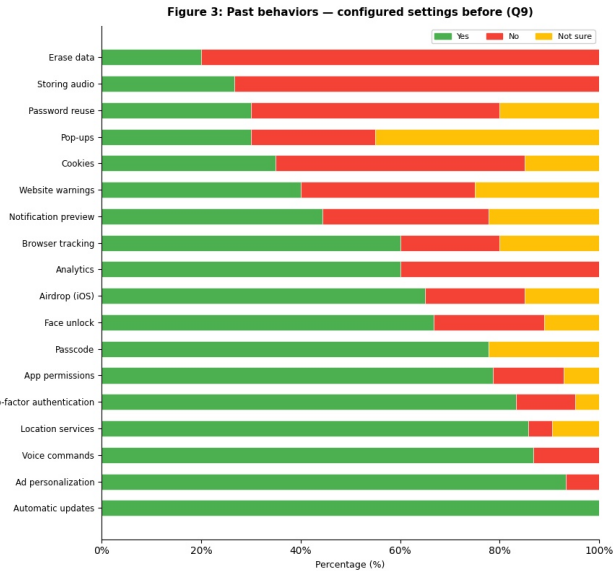


Figure 3: Participants' past configuration of smartphone privacy and security settings (Q9).

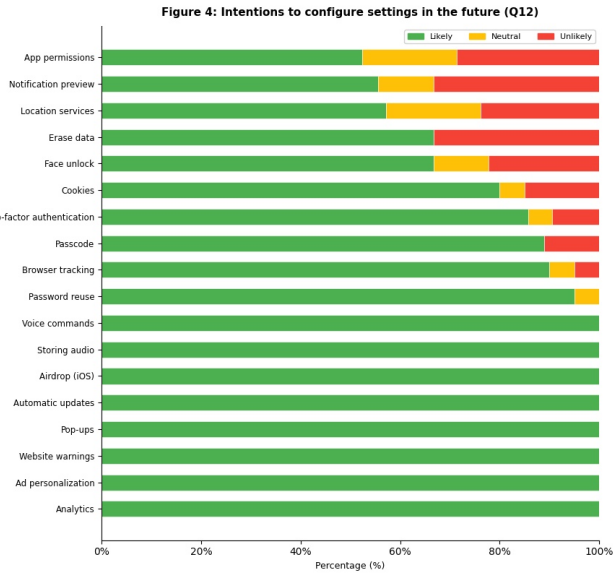


Figure 4: Participants' future intentions to configure smartphone privacy and security settings (Q12).

Participants most frequently reported configuring automatic updates, ad personalization, voice commands, location services, two-factor authentication, app permissions, passcodes, and face unlock. These settings are commonly encountered in everyday smartphone use and are often connected to account protection, app access, and device security.

In contrast, fewer participants had configured settings such as erase data, storing audio, password reuse, pop-ups, cookies, and

website warnings. These settings may be less visible, less clearly explained, or less directly connected in users' minds to everyday privacy and security protection.

Despite this, participants expressed strong intentions to configure many settings in the future, especially analytics, ad personalization, website warnings, pop-ups, automatic updates, AirDrop, storing audio, voice commands, password reuse, browser tracking, passcodes, and two-factor authentication. Even settings with low past configuration showed higher future willingness.

Consistent with Frik et al. [30], these findings suggest a gap between past behavior and future intention. However, in the Saudi context, this gap may reflect more than individual motivation. Users may be willing to improve their privacy and security practices, but need clearer Arabic explanations, better discoverability, and more context-aware guidance that connects each setting to familiar risks such as account access, location exposure, photos, and personal data protection.

5.6 Perceived Difficulty of Changing the Settings

To answer RQ4, we analyzed participants' responses about the perceived difficulty of changing the settings (Q7), which are summarized in Figure 5. In general, more than 72% of participants across all settings expected that it would be easy for them to configure the settings, consistent with the general finding that most users expect privacy settings to be easy to configure [30].

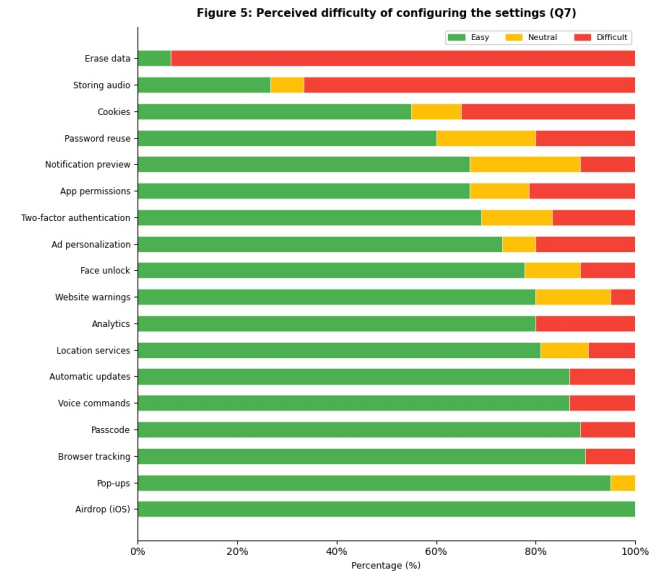


Figure 5: Perceived difficulty of configuring the settings (Q7).

No participants at all found it difficult to configure *AirDrop/Nearby Share* or block *Pop-ups* (0% difficult). Only about 11% of participants expected difficulties with *Passcode* and *Notification Preview*, and 13.3% with *Automatic Updates* and *Voice Commands*. Only a few participants expected difficulties with *Location Services* (9.5%), *Face Unlock* (11.1%), and *Two-Factor Authentication* (16.7%).

Regarding tracking technologies, less than 20% of participants expected difficulties with disabling *Browser Tracking* (10%), *Ad Personalization* (20%), *Analytics* (20%), and *Voice Commands* (13.3%). *Website Warnings* and *AirDrop* were among the most accessible settings for Saudi users.

Among the settings that the least number of users expected to be easy to configure, 66.7% of participants anticipated difficulties with opting out of *Storing Audio* recordings of voice assistant interactions (mean = 4.33), and 93.3% found *Erase Data* to be the most difficult setting by far (mean = 6.00). *Cookies* also showed notable difficulty, with 35% of participants finding it challenging (mean = 3.70). *App Permissions* (21.4%) and *Password Reuse Detection* (20%) also showed moderate anticipated difficulty.

5.6.1 Reasons for the Perceived Difficulty. We asked participants why they believed it would be difficult to configure the settings (Q8). Their responses were thematically coded and fell into three main groups: *Knowledge Gap*, *Unclear Interface or Instructions*, and *App Constraints*. Because some participants provided multiple reasons per setting, and for up to three different settings, we report occurrences (the number of times the reason was mentioned) instead of the number of people who mentioned it. This approach follows the qualitative analysis procedure used in Friik et al. [30].

Knowledge Gap was the most frequently cited reason, mentioned across multiple settings. For *Erase Data*, Knowledge Gap appeared 10 times out of 15 responses, indicating that most participants were simply unaware this feature existed or how to activate it. For *Storing Audio*, Knowledge Gap (5 times) combined with Navigation & Knowledge Issues (3 times) and Unclear (3 times) accounted for all reported difficulties, suggesting participants did not know where to locate this setting within their operating system. *Password Reuse Detection* also showed a mix of Navigation & Knowledge Issues (5 times), App Constraints (4 times), and Knowledge Gap (3 times), reflecting that participants were unsure whether this feature existed and where to find it.

Unclear responses (total 33 occurrences across all settings) reflected confusion about the setting's purpose, function, or method of configuration. This was prominent for *Two-Factor Authentication* (6 occurrences), *Passcode* (2 occurrences), and *Notification Preview* (2 occurrences), where participants were uncertain about the configuration process even when they knew the setting existed.

App Constraints (total 15 occurrences) referred to technical limitations where participants believed the setting was unavailable on their specific device model, operating system version, or region. This was notably reported for *Location Services* (5 occurrences) and *Password Reuse* (4 occurrences), suggesting some participants could not configure these settings due to platform or regional restrictions.

Notably, the majority of responses across settings were coded as *No Difficulty* (131 occurrences out of 363 total setting-level responses), consistent with the quantitative finding that most participants expected configuration to be easy [30]. Note that occurrences exceed the number of participants (N = 121) because each participant responded about up to three different settings. This pattern was particularly strong for *Automatic Updates* (14 out of 15 responses), *Voice Commands* (13/15), *Analytics* (11/15), and *Ad Personalization* (10/15), where participants reported no anticipated difficulty.

The distribution of difficulty reasons was not the same across settings. For most settings, difficulties were primarily associated with Knowledge Gap and navigation issues. In contrast, *Erase Data* showed an unusual pattern where 93.3% of participants found it difficult yet no participants cited App Constraints — suggesting the difficulty stems from unfamiliarity with an obscure but technically accessible feature, rather than device or platform limitations.

5.6.2 Factors Influencing Perceived Difficulty. To answer RQ6c, we estimated ordered logistic regression models predicting perceived difficulty of configuring settings (Q7; 1=Very Easy to 7=Very Difficult), following the analytical approach of Friik et al. [30] (see Table 7 in Appendix B for full results). Model 1 included demographic predictors only ($\chi^2(1) = 7.14, p = .008$); Model 2 added smartphone-related covariates ($\chi^2(8) = 18.04, p = .021$).

In Model 1, older age was a significant positive predictor of perceived difficulty ($b = 0.33$, 95% CI [0.02, 0.65], OR = 1.39, $p < .05$), indicating that each unit increase in age group was associated with 39% higher odds of reporting greater configuration difficulty. This replicates Friik et al.'s finding that older adults perceive privacy settings as harder to configure [30], and likely reflects lower exposure to evolving Arabic privacy terminology in smartphone interfaces.

In Model 2, smartphone ease of use was a significant positive predictor ($b = 0.28$, 95% CI [0.09, 0.47], OR = 1.32, $p < .01$), suggesting that participants who find general smartphone tasks easier also perceive privacy settings as easier to configure. The age effect was attenuated once smartphone ease of use was included, indicating that general digital fluency partially mediates the age–difficulty relationship. Technical background showed a marginal negative effect ($b = -0.64$, 95% CI [-1.35, 0.07], OR = 0.53, $p < .1$), consistent with the expectation that technical expertise reduces perceived configuration difficulty [30]. No significant effects were found for gender, education, income, iOS vs. Android, or OS experience duration, which may reflect the demographic homogeneity of our Saudi sample (95% with >2 years OS experience).

The overall model fit was weaker than in Friik et al. ($\chi^2 M1 = 7.14$, ns vs. 44.50^{***} , $p < .001$), consistent with reduced demographic variance in our sample rather than model invalidity. These findings partially replicate Friik et al. [30] while extending them to a Saudi Arabic-speaking context in which linguistic barriers and device familiarity patterns differ substantially from those observed in the US.

5.7 Perceived Impact of Settings on User Experience

To answer RQ5, participants were asked how configuring privacy and security settings would affect their overall smartphone experience (Q11).

Figure 6: Perceived impact of settings on user experience (Q11)

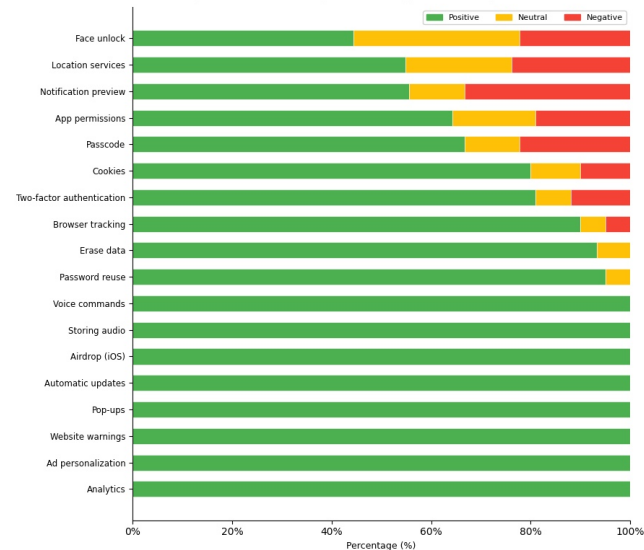


Figure 6: Perceived impact of settings on overall user experience (Q11).

As shown in Figure 6, participants generally perceived configuring privacy and security settings as having a positive impact on their overall smartphone experience. *Two-Factor Authentication (2FA)* was associated with the most positive perception, with 80% of participants indicating that enabling it would improve their experience, and only a small proportion reporting neutral (8%) or negative (12%) expectations.

App Permissions were also viewed positively by the majority of participants (65%), although a noticeable proportion reported neutral (18%) or negative (18%) expectations. This suggests that while users recognize the importance of controlling app access, some may associate it with potential inconvenience or reduced functionality.

Location Access showed the lowest positive perception among the three settings, with 55% of participants reporting a positive impact, while 22% selected neutral and another 22% anticipated a negative impact. This indicates that restricting location services may be perceived as affecting usability or limiting app features.

Overall, these findings suggest that while users generally acknowledge the benefits of privacy and security settings, their perceived impact on user experience varies depending on the type of setting. In particular, security-focused features such as 2FA are viewed more favorably, whereas settings that directly affect app functionality, such as location access and permissions, may introduce concerns about usability.

5.8 Effectiveness of Settings Against Risks

To answer RQ5b, we asked participants to rate how effective each smartphone privacy and security setting is in protecting them from privacy and security risks (Q13). Figure 7 presents the mean effectiveness scores for the 18 settings. Overall, participants viewed most settings as effective, as nearly all settings scored above the neutral midpoint.

The strongest perceived protection was associated with website warnings ($M = 6.70$), voice commands ($M = 6.60$), analytics ($M = 6.27$), automatic updates ($M = 6.20$), ad personalization ($M = 6.13$), password reuse detection ($M = 6.10$), and storing audio controls ($M = 6.07$). Pop-up blocking ($M = 6.00$) and two-factor authentication ($M = 5.88$) were also rated highly. This may reflect participants' greater trust in settings that appear directly related to warnings, account protection, tracking control, and data collection [20, 30, 72].

Other settings received positive but lower effectiveness scores, including browser tracking controls ($M = 5.70$), erase data ($M = 5.67$), AirDrop ($M = 5.65$), cookies ($M = 5.50$), notification preview ($M = 5.44$), passcode ($M = 5.33$), location services ($M = 5.17$), and app permissions ($M = 5.05$). Although these settings were still viewed as useful, the lower scores suggest less confidence in their protective role compared with the highest-rated settings.

Face unlock received the lowest effectiveness score ($M = 4.33$), making it the only setting close to the neutral midpoint. This may indicate uncertainty among participants about whether biometric authentication provides strong privacy or security protection, consistent with prior work showing that users may differ in how they perceive the security and usability of authentication mechanisms [22, 32].

Overall, participants generally perceived smartphone privacy and security settings as useful protective tools, but their confidence varied across settings. This variation suggests that Saudi users may recognize the value of privacy and security controls while still needing clearer explanations of what each setting actually protects [8, 20, 28, 30].

5.9 Perceived Worry About the Risks

To answer RQ5, we asked participants to indicate their level of agreement with statements describing common privacy and security risks related to smartphone use (Q15). Overall, participants expressed concern about most of the risks included in the survey. The strongest concerns were related to hacking or malware attacks, unauthorized access to smartphones, phishing and scam attacks, and disclosure of personal information. These risks reflect participants' concerns about direct security threats, account compromise, and exposure of sensitive data.

Participants also showed concern about data collection by companies and governments, indicating awareness of broader privacy risks related to tracking, surveillance, and personal data use. In the Saudi context, this concern may be connected to the central role of smartphones in accessing banking services, digital identity platforms, government applications, and social communication. As a result, risks involving personal data, accounts, and location information may feel especially important to users.

In comparison, risks such as unsolicited advertising and objectionable content appeared to be of lower concern than risks involving unauthorized access, scams, or personal data disclosure. This suggests that participants were more concerned about risks that could lead to direct harm, loss of control, or exposure of sensitive information than about risks perceived as annoying or less personally damaging.

Overall, these findings suggest that Saudi smartphone users are concerned about both security-related threats and privacy-related

Mean Scores per Privacy Setting (Q5-Q13)
Green = positive/high, Red = negative/low

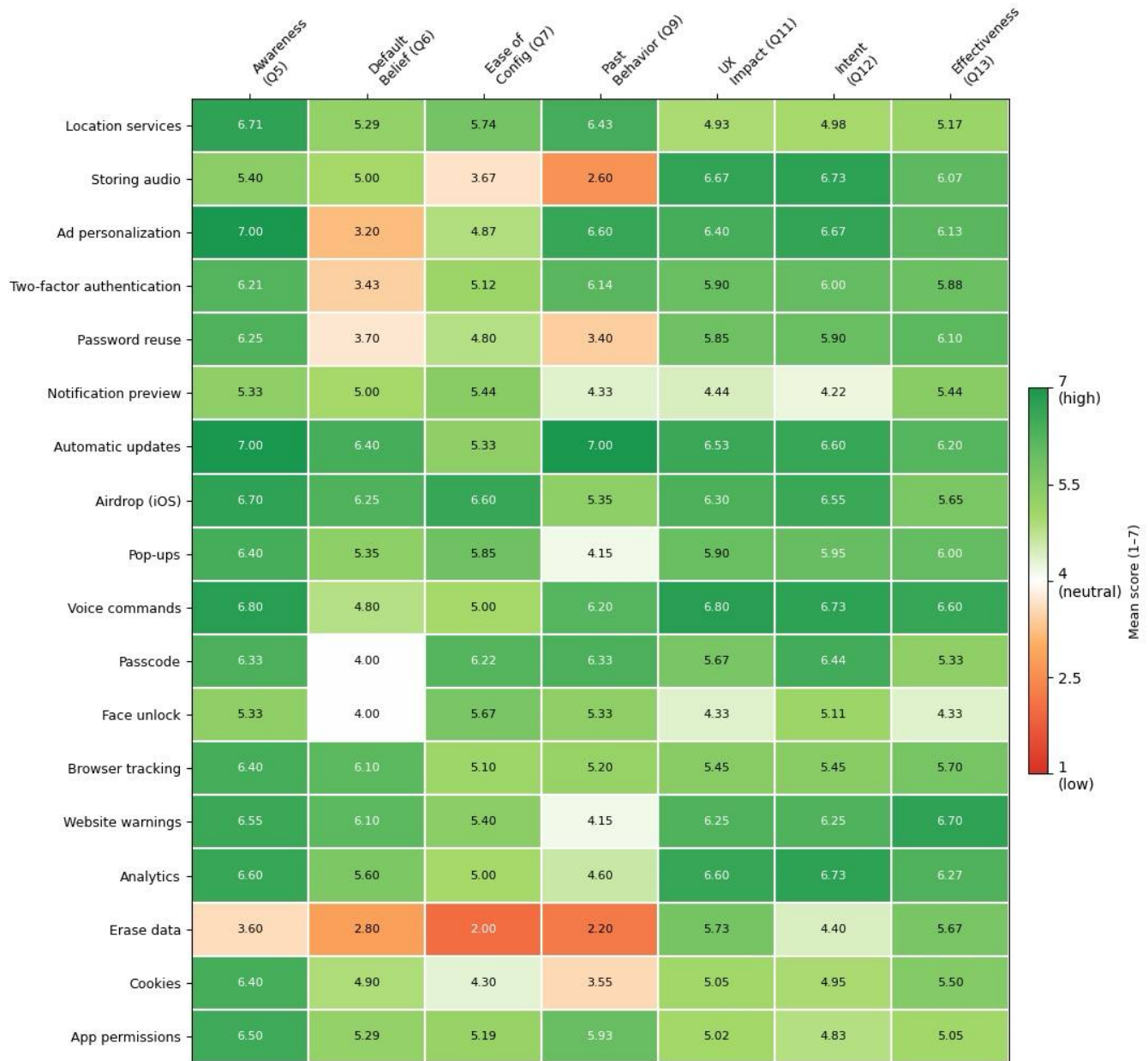


Figure 7: Mean effectiveness scores across the 18 smartphone privacy and security settings (Q13). Green indicates higher perceived effectiveness, and red indicates lower perceived effectiveness.

data practices. Consistent with RQ5, users' perceptions of smartphone privacy and security settings appear to be shaped by the risks they consider most serious, particularly unauthorized access, online fraud, personal data disclosure, and risks connected to everyday digital services in Saudi Arabia.

6 Discussion and Conclusions

The findings indicate that Saudi smartphone users showed relatively high awareness of many privacy and security settings, yet this did not consistently translate into configuration behavior. Participants recognized common controls such as automatic updates, location

services, and two-factor authentication, but remained uncertain about default configurations and less visible settings. Similar to Frik et al. [30], these findings confirm an awareness–behavior gap [14, 19, 39, 54, 57, 58, 70], shaped in the Saudi context by cultural expectations, institutional trust, and Arabic technical terminology.

A central contribution is showing that privacy perceptions in Saudi Arabia are culturally situated. Prior work frames privacy as individual data control [6, 30, 53], whereas in Gulf societies privacy extends to family reputation, social visibility, and collective expectations [2–5]. This aligns with collectivist cultural frameworks [33, 34, 66], and helps explain participants’ heightened concern about location tracking, photo access, and notification exposure.

Institutional trust also shaped responses: participants linked privacy risks to banking, government platforms, and digital identity rather than commercial tracking alone [16, 21, 30, 62?, 63]. Language emerged as a parallel barrier — uncertainty around settings such as erase data, storing audio, and cookies reflects the difficulty of interpreting Arabic translations of technical privacy terms [8, 20, 27, 30, 38, 44, 49, 72].

Taken together, the gap between awareness and configuration in Saudi Arabia is shaped by three intersecting dimensions: collectivist privacy norms, institutional trust ecosystems, and Arabic technical interpretation. This is supported by contextual privacy theory [53], cross-cultural privacy research [34, 43, 64], work on privacy decision-making [41, 42], and prior work on privacy behavior [14, 54, 70]. Detailed comparisons with Frik et al. [30] across awareness, past behavior, perceived difficulty, regression results, gender differences, and design implications are provided in Tables A–F (Appendix B).

Privacy interfaces for Arabic-speaking users should be context-aware rather than relying on direct translation, with settings linked to realistic Saudi scenarios [8, 18, 20, 28, 31, 49, 50, 55, 71].

6.1 Socio-demographic Differences

Female participants (57.5%) showed higher expressed concern but comparable configuration behavior to male participants, consistent with Frik et al. [30] and broader literature [14, 54]. Younger participants (18–34 years, 82.5%) configured more settings, reflecting greater digital familiarity. Participants with technical backgrounds perceived less difficulty, while non-technical users struggled particularly with App Permissions, supporting evidence that poor interface design disproportionately affects users with lower digital literacy [30, 59]. Regional variation was observed, with the Western Region overrepresented (60%), which limits generalizability; future work should use stratified sampling.

6.2 Recommendations

Smartphone operating systems should improve the discoverability of less familiar settings, especially erase data, storing audio, and password reuse detection, through context-sensitive prompts and periodic privacy checkups [23, 30, 35, 47, 52, 59].

We propose a culturally localized privacy framework for Arabic-speaking users with three layers: clear Arabic labels, short explanations of data collected and risks reduced, and Saudi-relevant examples linking settings to photos, location, banking, and family privacy [2–5, 34, 46, 49, 53].

At the policy level, SDAIA/NDMO should consider language clarity standards for Arabic privacy terminology in digital services; recent PDPL implementation analysis suggests this gap remains unaddressed [9]. Awareness campaigns by the National Cybersecurity Authority should be specific, actionable, and tailored to users’ actual risk contexts [61], focusing on practical scenarios such as managing location access, app permissions, and account protection in Arabic.

Users’ willingness to engage with settings is also shaped by their perception of data value and platform trustworthiness [29, 36]. Privacy interfaces should therefore clarify whether a risk originates from apps, platforms, websites, or the device itself [6, 7, 20, 44, 56, 67]. Future research should examine how Saudi users across regions and skill levels understand privacy settings and Arabic terminology to support culturally grounded interface design [1, 10, 12, 25, 43, 56, 64].

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References

- [1] 2025. Unplug, Mute, Avoid. arXiv preprint arXiv:2508.04202.
- [2] Norah Abokhodair et al. 2016. Privacy and Twitter in Qatar. In *Proceedings of WebSci*.
- [3] Norah Abokhodair, Sofiane Abbar, Sarah Vieweg, and Yelena Mejova. 2017. Privacy and Social Media Use in the Arabian Gulf. *Journal of Web Science* 3 (2017).
- [4] Norah Abokhodair and Amy Hodges. 2019. Toward a Transnational Model of Social Media Privacy. *New Media & Society* 21, 5 (2019), 1105–1120.
- [5] Norah Abokhodair and Sarah Vieweg. 2016. Privacy & Social Media in the Context of the Arab Gulf. In *Proceedings of DIS*. 672–683.
- [6] Alessandro Acquisti, Laura Brandimarte, and George Loewenstein. 2015. Privacy and Human Behavior in the Age of Information. *Science* 347, 6221 (2015), 509–514.
- [7] Alessandro Acquisti, Curtis Taylor, and Liad Wagman. 2016. The Economics of Privacy. *Journal of Economic Literature* 54, 2 (2016), 442–492.
- [8] Anne Adams and Martina Angela Sasse. 1999. Users Are Not the Enemy. *Commun. ACM* 42, 12 (1999), 40–46.
- [9] Esra Alashwali and Abdulrahman Alhuzali. 2026. One Year After the PDPL. *International Journal of Information Security and Privacy* (2026).
- [10] Esra Alashwali and Lorrie Faith Cranor. 2024. Privacy Perceptions and Behaviors of Google Personal Account Holders in Saudi Arabia. In *HCHI*. Springer.
- [11] Nawaf Alhalafi and Prakash Veeraraghavan. 2023. Exploring the Challenges and Issues in Adopting Cybersecurity in Saudi Smart Cities. *Smart Cities* 6, 3 (2023), 1523–1544.
- [12] Noof D. Alotaibi, Manal Abulkhair, and Manal Bayousef. 2026. Comprehensive Privacy Awareness Framework (CPAF). *JTAER* 21, 2 (2026).
- [13] Abdullah Alzubaidi. 2021. Cybercrime Awareness Among Saudi Nationals. *Data in Brief* 39 (2021).
- [14] Susanne Barth and Menno D. T. de Jong. 2017. The Privacy Paradox. *Telematics and Informatics* 34, 7 (2017), 1038–1058.
- [15] Nicolas Borgert et al. 2023. Home Is Where the Smart Is. In *Proceedings of CHI*. ACM.
- [16] Communications, Space and Technology Commission (CST). 2025. *Saudi Internet Report 2024*. Technical Report. CST, Riyadh, Saudi Arabia.
- [17] Lisa Y. Connolly, Michael Lang, and Jason Giboney. 2025. A Study of the Privacy Paradox Amongst Young Adults in the UAE. *Telematics and Informatics Reports* 19 (2025), 100248.
- [18] Lorrie Faith Cranor. 2005. *Security and Usability*. O’Reilly Media, Sebastopol, CA, USA.
- [19] Robert E. Crossler and France Bélanger. 2019. Why Would I Use Location Protective Settings on My Smartphone? *Information Systems Research* 30, 3 (2019), 995–1006.
- [20] Sauvik Das et al. 2022. The Security & Privacy Acceptance Framework. *Foundations and Trends in Privacy and Security* 5, 1–2 (2022), 1–143.
- [21] Digital Government Authority. 2025. *Digital Government Strategy 2023–2030*. Technical Report. DGA, Riyadh, Saudi Arabia.

- [22] Serge Egelman et al. 2014. Are You Ready to Lock?. In *Proceedings of ACM CCS*. 2019, 2 (2019), 209–227.
- [23] Serge Egelman, Adrienne Porter Felt, and David Wagner. 2013. Choice Architecture and Smartphone Privacy: There's a Price for That. In *The Economics of Information Security and Privacy*. Springer, Berlin, Germany, 211–236.
- [24] Matthew Evans et al. 2016. Human Behaviour as an Aspect of Cyber Security Assurance. *Security and Communication Networks* (2016).
- [25] Awais Farooq et al. 2024. Exploring Social Media Privacy Concerns. *IEEE Access* 12 (2024), 147087–147105.
- [26] Adrienne Porter Felt et al. 2011. Improving Android Permissions. In *USENIX Security*.
- [27] Adrienne Porter Felt, Erika Ha, Serge Egelman, Ariel Haney, Erika Chin, and David Wagner. 2012. Android Permissions: User Attention, Comprehension, and Behavior. In *Proceedings of the Eighth Symposium on Usable Privacy and Security (SOUPS)*.
- [28] Ana Ferreira, Lynne Coventry, and Gabriele Lenzini. 2014. Principles of Usable Security. *Information* 5, 2 (2014), 319–343.
- [29] Peter Fleming et al. 2021. The Role of Personal Data Value. *PLOS ONE* 16, 7 (2021).
- [30] Alisa Frik, Juvan Kim, Julia R. Sanchez, and Junliang Ma. 2022. Users' Expectations About and Use of Smartphone Privacy and Security Settings. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA. <https://doi.org/10.1145/3491102.3517504>
- [31] Marthie Grobler, Raj Gaire, and Surya Nepal. 2021. User, Usage and Usability. *Frontiers in Big Data* 4 (2021).
- [32] Marian Harbach et al. 2014. It's a Hard Lock Life. In *Proceedings of SOUPS*.
- [33] Geert Hofstede. 1983. National Cultures in Four Dimensions. *International Studies of Management & Organization* 13, 1–2 (1983), 46–74.
- [34] Geert Hofstede, Gert Jan Hofstede, and Michael Minkov. 2010. *Cultures and Organizations* (3rd ed.). McGraw-Hill.
- [35] Iulia Ion et al. 2015. No More Guestimates. In *SOUPS*.
- [36] Qatrunnada Ismail et al. 2017. To Permit or Not to Permit. *PoPETS 2017*, 4 (2017), 119–137.
- [37] Patrick Gage Kelley et al. 2013. Privacy as Part of the App Decision-Making Process. In *Proceedings of CHI*.
- [38] Patrick Gage Kelley, Sunny Consolvo, Lorrie Faith Cranor, Jaeyeon Jung, Norman Sadeh, and David Wetherall. 2012. A Conundrum of Permissions: Installing Applications on an Android Smartphone. In *Financial Cryptography and Data Security*. Springer, Berlin, Germany, 68–79.
- [39] Paul E. Ketelaar and Maaik Van Balen. 2018. The Smartphone as Your Follower. *Computers in Human Behavior* 78 (2018), 174–182.
- [40] Jvision2030 Kingdom of Saudi Arabia. [n. d.]. Saudi Vision 2030.
- [41] Bart P. Knijnenburg et al. (Eds.). 2022. *Modern Socio-Technical Perspectives on Privacy*. Springer.
- [42] Bart P. Knijnenburg and Alfred Kobsa. 2013. Making Decisions About Privacy. *ACM TIS* 3, 3 (2013).
- [43] Yuan Li. 2022. Cross-Cultural Privacy Differences. In *Modern Socio-Technical Perspectives on Privacy*. Springer.
- [44] Jialiu Lin, Shahriyar Amini, Jason I. Hong, Norman Sadeh, Janne Lindqvist, and Joy Zhang. 2012. Expectation and Purpose: Understanding Users' Mental Models of Mobile App Privacy Through Crowdsourcing. In *Proceedings of the ACM Conference on Ubiquitous Computing*. 501–510.
- [45] Jialiu Lin, Bin Liu, Norman Sadeh, and Jason I. Hong. 2014. Modeling Users' Mobile App Privacy Preferences: Restoring Usability in a Sea of Permission Settings. In *Tenth Symposium on Usable Privacy and Security (SOUPS)*.
- [46] Heather Richter Lipford et al. 2008. Understanding Privacy Settings in Social Networking Sites. CHI Workshop.
- [47] Bin Liu et al. 2016. Follow My Recommendations: A Personalized Privacy Assistant for Mobile App Permissions. In *Twelfth Symposium on Usable Privacy and Security (SOUPS)*. 27–41.
- [48] Mary Madden. 2017. Privacy, Security, and Digital Inequality. *Data & Society* (2017).
- [49] Aaron Marcus and Eugene W. Gould. 2000. Crosscurrents. *Interactions* 7, 4 (2000), 32–46.
- [50] Philipp K. Masur and Sabine Trepte. 2017. *Privacy Online*. Springer.
- [51] Nicholas Micallef et al. 2015. Why Aren't Users Using Protection?. In *Proceedings of MobileHCL*.
- [52] Jakob Nielsen. 1994. *Usability Engineering*. Academic Press.
- [53] Helen Nissenbaum. 2010. *Privacy in Context*. Stanford University Press.
- [54] Patricia A. Norberg et al. 2007. The Privacy Paradox. *Journal of Consumer Affairs* 41, 1 (2007), 100–126.
- [55] Donald A. Norman. 2013. *The Design of Everyday Things*. Basic Books.
- [56] OECD. 2022. *Building Trust to Reinforce Democracy*. OECD, Paris.
- [57] Yong Jin Park. 2013. Digital Literacy and Privacy Behavior Online. *Communication Research* 40, 2 (2013), 215–236.
- [58] Yong Jin Park, Scott W. Campbell, and Nojin Kwak. 2012. Affect, Cognition and Reward. *Computers in Human Behavior* 28, 3 (2012), 1019–1027.
- [59] Kopo M. Ramokapane, Awais C. Mazeli, and Awais Rashid. 2019. Skip, Skip, Skip, Accept!!!: A Study on the Usability of Smartphone Manufacturer Provided Default Features and User Privacy. *Proceedings on Privacy Enhancing Technologies* 2019, 2 (2019), 209–227.
- [60] Elissa M. Redmiles et al. 2017. Where Is the Digital Divide?. In *Proceedings of CHI*.
- [61] Elissa M. Redmiles et al. 2020. A Comprehensive Quality Evaluation of Security and Privacy Advice on the Web. In *Proceedings of USENIX Security*.
- [62] Saudi Data and Artificial Intelligence Authority (SDAIA). 2025. *Personal Data Protection Law and Implementing Regulations*. Technical Report. SDAIA, Riyadh, Saudi Arabia.
- [63] Scott J. Savage and Donald M. Waldman. 2015. Privacy Tradeoffs in Smartphone Applications. *Economics Letters* 137 (2015), 171–175.
- [64] Sabine Trepte et al. 2017. A Cross-Cultural Perspective on the Privacy Calculus. *Social Media + Society* 3, 1 (2017).
- [65] Sabine Trepte, Tobias Dienlin, and Leonard Reinecke. 2014. Risky Behaviors: How Online Experiences Influence Privacy Behaviors. (2014).
- [66] Harry C. Triandis. 1995. *Individualism and Collectivism*. Westview Press.
- [67] United Nations. 2024. *United Nations E-Government Survey 2024*. United Nations, New York.
- [68] Daniele Vecchiato et al. 2015. A Security Configuration Assessment for Android Devices. In *Proceedings of ACM SAC*.
- [69] Michel Walrave et al. 2012. Connecting and Protecting? *Cyberpsychology* 6, 1 (2012).
- [70] Mike Z. Yao et al. 2007. Predicting User Concerns About Online Privacy. *JASIST* 58, 5 (2007), 710–722.
- [71] Ka-Ping Yee. 2004. Aligning Security and Usability. *IEEE Security & Privacy* 2, 5 (2004), 48–55.
- [72] Yucheng Zhou, Alexander Raake, Ting Xu, and Xiaofeng Zhang. 2017. Users' Perceived Control, Trust and Expectation on Privacy Settings of Smartphone. In *International Symposium on Cyberspace Safety and Security*. 427–441.

A Descriptive Results and Logistic Regression Analysis

Table 3: Participants’ overall concern about privacy and security risks (Q15 in the survey; 7-point Likert scale).

Measure	Mean	Unconcerned (%)	Neutral (%)	Concerned (%)
Overall concern about privacy and security risks	5.76	5.88	9.24	84.87

Table 4: Logistic Regression – Past Behavior (Q9) (N = 119)

	(1)	(2)	(3)
Age	-0.3620 [-0.86, 0.13]	-0.2097 [-0.81, 0.39]	-0.2119 [-0.85, 0.42]
Female (Male omitted)	-1.2786 [-2.86, 0.31]	-1.6293 ⁺ [-3.36, 0.10]	-1.6518 ⁺ [-3.40, 0.10]
Education	0.0130 [-0.55, 0.57]	0.0410 [-0.59, 0.67]	0.0283 [-0.61, 0.66]
Income	0.1788 [-0.29, 0.65]	0.1977 [-0.32, 0.71]	0.2506 [-0.29, 0.79]
iOS users (Android omitted)		1.6472* [0.24, 3.05]	1.5136* [0.06, 2.97]
Duration of experience with OS		0.1308 [-1.29, 1.56]	0.1309 [-1.32, 1.58]
Smartphone ease of use		0.0825 [-0.32, 0.48]	0.0618 [-0.35, 0.48]
Technical background		0.7505 [-0.65, 2.15]	0.8324 [-0.63, 2.30]
Perceived difficulty to configure settings			0.0119 [-0.66, 0.68]
Perceived negative impact on UX			0.2547 [-0.57, 1.08]
χ^2	5.32	11.95	12.37

95% confidence intervals in brackets ⁺ $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$ **Table 5: Ordered Logistic Regression – Intent to Configure (Q12) (N = 119)**

	(1)	(2)	(3)
Age	-0.0576 [-0.37, 0.25]	-0.0584 [-0.39, 0.28]	0.0596 [-0.32, 0.44]
Female (Male omitted)	-0.2031 [-0.91, 0.51]	-0.2067 [-0.96, 0.55]	-0.2317 [-1.06, 0.59]
Education	0.0152 [-0.30, 0.33]	-0.0075 [-0.34, 0.33]	0.1559 [-0.20, 0.51]
Income	0.4758*** [0.20, 0.75]	0.4799*** [0.20, 0.76]	0.2677** [-0.05, 0.58]
iOS users (Android omitted)	-0.6503 ⁺ [-1.40, 0.10]	0.1577 [-0.70, 1.02]	
Duration of experience with OS	-0.5222 [-1.33, 0.29]	-0.5358 [-1.35, 0.27]	
Smartphone ease of use	-0.1818 ⁺ [-0.38, 0.01]	-0.0114 [-0.24, 0.21]	
Technical background	0.4823 [-0.24, 1.21]	0.4117 [-0.40, 1.23]	
Perceived difficulty to configure settings		-0.0286 [-0.40, 0.34]	
Past behavior with configuring settings		-0.5459 [-1.77, 0.68]	
Perceived negative impact on UX		-2.2856*** [-2.89, -1.68]	
χ^2	12.74*	23.98**	112.17***

95% confidence intervals in brackets ⁺ $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

B Comparison Tables: Present Study vs. Friik et al. [30]

The following six tables compare findings from the present Saudi replication study against Friik et al. [30], organized around three analytical dimensions: (1) Collectivist Privacy [3, 5, 64]; (2) Linguistic Opacity [1, 43, 49]; and (3) the Trust Ecosystem [21, 56, 62].

Table 6: Ordered Logistic Regression – Perceived Difficulty (Q7) (N = 119)

	(1)	(2)
Age	0.3321*	0.1823
	[0.02, 0.65]	[-0.16, 0.52]
Female (Male omitted)	-0.0840	0.2507
	[-0.76, 0.60]	[-0.47, 0.97]
Education	-0.2094	-0.2026
	[-0.52, 0.10]	[-0.53, 0.13]
Income	0.1792	0.2573**
	[-0.08, 0.44]	[-0.02, 0.53]
iOS users (Android omitted)		-0.5699
		[-1.32, 0.18]
Duration of experience with OS		0.0334
		[-0.69, 0.76]
Smartphone ease of use		0.2787**
		[0.09, 0.47]
Technical background		-0.6380**
		[-1.35, 0.07]
χ^2	7.14	18.04*

95% confidence intervals in brackets * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ **** $p < 0.001$

Table 7: Ordered Logistic Regression – Negative UX Impact (Q11) (N = 119)

	(1)	(2)
Age	0.1305	0.1290
	[-0.18, 0.44]	[-0.21, 0.47]
Female (Male omitted)	0.2457	0.2815
	[-0.47, 0.96]	[-0.48, 1.04]
Education	0.1617	0.2211
	[-0.16, 0.49]	[-0.12, 0.56]
Income	-0.4280**	-0.4420**
	[-0.70, -0.15]	[-0.72, -0.16]
iOS users (Android omitted)		0.8571*
		[0.11, 1.61]
Duration of experience with OS		0.1051
		[-0.58, 0.79]
Smartphone ease of use		0.2082*
		[0.01, 0.41]
Technical background		-0.3524
		[-1.09, 0.38]
χ^2		11.61

95% confidence intervals in brackets * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ **** $p < 0.001$

Saudi users showed notably higher awareness of Ad personalization (100% vs <50%), Analytics (86.7%), and Password reuse (75%) than Frik et al.'s US sample [56?], likely due to social media engagement and digital literacy initiatives. Notification preview awareness was lower (44.4% vs ~75%), possibly reflecting collectivist norms of shared device use [5, 64]. Erase data was similarly low in both studies (~20%) [30, 43].

The most significant finding is the high 2FA configuration rate (83.3%), explained by mandatory requirements of Saudi government platforms such as Absher and Nafath [21, 62]. Ad personalization configuration was also high (93.3%) [5, 56]. Storing audio (26.7%) and Erase data (20%) remained low, consistent with the privacy paradox among Gulf youth [14, 17, 54].

The most striking finding is the reversal for Erase data: easy for ~97% of US participants but difficult for 93.3% of Saudi participants, likely due to ambiguity of “ ” in Arabic [43, 49]. Storing audio was also more difficult in the Saudi sample (66.7% vs 19–25%) [1]. AirDrop and Pop-ups showed 0% difficulty, reflecting social familiarity [3, 5, 56].

Age and technical background showed consistent effects across both studies [13, 48]. OS duration did not replicate, reflecting a ceiling effect (95% with >2 years experience). The weaker model fit (χ^2 M1=7.14 vs 44.50****) reflects demographic homogeneity rather than model invalidity [1, 60, 64].

Table 8: OLS Regression — General Privacy & Security Concern (Q15) (N = 119)

	(1)	(2)
Age	0.0865 [-0.14, 0.31]	0.1466 [-0.08, 0.38]
Female (Male omitted)	0.2074 [-0.29, 0.71]	0.0262 [-0.49, 0.54]
Education	-0.0266 [-0.24, 0.19]	-0.0616 [-0.28, 0.16]
Income	0.3003** [0.11, 0.49]	0.2927** [0.10, 0.48]
iOS users (Android omitted)		-0.1195 [-0.64, 0.40]
Duration of experience with OS		0.1746 [-0.28, 0.63]
Smartphone ease of use		-0.1863** [-0.32, -0.05]
Technical background		0.3057 [-0.19, 0.80]
χ^2		10.37*

95% confidence intervals in brackets * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ **** $p < 0.001$

The gender-risk paradox identified in Frik et al. (women reporting less privacy concern despite greater exposure) did not replicate in Saudi Arabia (coef=0.21, ns). Saudi women’s privacy concerns appear more socially grounded — centered on location tracking, photo exposure, and family surveillance — consistent with collectivist privacy frameworks [4, 5, 25, 43, 56, 64, 70].

Four design recommendations emerge from the three cultural dimensions. The most impactful is co-designing Arabic privacy terminology with native users [43]. Integrating privacy checkups into government platforms (Nafath, Absher) capitalizes on existing institutional trust [21, 62]. Together these argue for culturally-situated privacy engineering for Arabic-speaking users [1, 20, 31].

Table A: Awareness of Settings (Q5)

Table 9: Awareness of Settings (Q5) – Present Study vs. Frik et al. [30]

Setting	Frik et al. (%)	Present Study (%)	Cultural Interpretation
Location services	> 75%	92.9%	Location privacy is socially significant in Saudi Arabia — tracking is conflated with social surveillance by family or state actors. High awareness reflects everyday sensitivity [16,17,60].
Ad personalization	< 50%	100%	All Saudi participants were aware, likely driven by high social media penetration and exposure to targeted advertising discussions in Arabic online communities [19,58].
Automatic updates	> 75%	100%	Full awareness — OS updates are perceived as routine digital maintenance in both contexts. No cultural gap observed.
Analytics	~ 50%	86.7%	Higher awareness may reflect heightened distrust of data collection by non-Saudi platforms, consistent with digital sovereignty discourse under Vision 2030 [23,24].
Notification preview	~ 75%	44.4%	Striking gap. In collectivist Saudi households where devices are shared, notification content may be routinely visible to family members, reducing perceived need to restrict previews. The Arabic term may also be unfamiliar [52,54,60].
Erase data	~ 20%	20.0%	Low in both studies. The Arabic translation “ ” is ambiguous — users may conflate it with a factory reset, reducing recognition of it as a configurable security feature [54,55].
Password reuse	< 50%	75.0%	Higher awareness than Frik et al. may reflect increased concern about account hijacking in Saudi digital banking and government portal contexts [20,25].

Table B: Past Configuration Behavior (Q9)

Table 10: Past Configuration Behavior (Q9) – Present Study vs. Friik et al. [30]

Setting	Friik et al.	Present Study (% Configured)	Cultural Interpretation
Two-factor authentication	Not majority	83.3%	HIGHEST GAP. 2FA is mandatory for Saudi government portals (Absher, Nafath), creating behavioral familiarity through institutional enforcement rather than user initiative [24,25].
Ad personalization	35–65%	93.3%	Highest behavioral gap. Saudi users widely disable ad tracking – possibly driven by awareness of commercial surveillance and privacy norms in a high social-media context [58,60].
Automatic updates	60–80%	100%	Full configuration – consistent with government push for digital infrastructure under Vision 2030 and mandatory OS update policies in enterprise contexts [23].
Voice commands	60–80%	86.7%	High – privacy of voice is culturally sensitive in Saudi multigenerational homes. Disabling “Hey Siri/Google” may reflect concern about ambient listening near family members [16,61].
Storing audio	> 80%	26.7%	Low despite awareness – may reflect privacy fatalism (“the company already has my data”) or low self-efficacy, consistent with the privacy paradox observed among Gulf youth [39,57].
Pop-ups	35–65%	30.0%	Low – Arabic browser interfaces may not clearly expose pop-up blocking settings, consistent with linguistic usability barriers [54].
Erase data	> 80%	20.0%	Low in both studies – unfamiliarity with the security feature persists across cultural contexts, suggesting this is a universal discoverability failure [1,8].

Table C: Perceived Difficulty (Q7)

Table 11: Perceived Difficulty (Q7) – Present Study vs. Frik et al. [30]

Setting	Frik et al. (% Difficult)	Present Study (% Difficult)	Interpretation
Erase data	~ 3% (easy for 97%)	93.3%	Striking reversal: ambiguity of Arabic translation “ ” conflated with factory reset [43,49].
Storing audio	19–25%	66.7%	Higher difficulty in Saudi sample due to unfamiliar terminology and navigation issues.
AirDrop/Nearby Share	Not reported	0%	Social familiarity and frequent use in Saudi context [3,5,56].
Block Pop-ups	Not reported	0%	Perceived as trivial; browser settings accessible.

Table D: Regression Results – Perceived Difficulty (Q7)

Table 12: Regression Results – Perceived Difficulty (Q7)

Predictor	Frik et al. [30]	Present Study	Cultural Interpretation
Age (older)	0.0155* M1 / 0.0134 ns M2	0.33* M1 / 0.18 ns M2	Partial. Age effect replicated across cultures. Older Saudi users have less exposure to evolving Arabic privacy terminology in OS updates, amplifying the knowledge gap [20,46,52].
Gender (Female)	-0.360+ M2	ns	Partial. No gender gap in Saudi sample. Unlike US findings, Saudi women are active smartphone users under Vision 2030 digital inclusion – gender digital divide differs from Western patterns [16,23,52].
Technical background	-0.406* M2	-0.638+ M2	Partial. Consistent direction – tech expertise reduces perceived difficulty regardless of cultural context. Interface complexity, not culture alone, is the primary barrier for tech-literate users [31,33,59].
OS duration	-0.453* M2	ns	NOT replicated. 95% of our sample had >2 years OS experience, eliminating variance. Arabic terminology barriers persist independently of OS familiarity [54,55].
iOS vs Android	ns	ns	Yes. No platform effect in either study. However, Arabic UI quality varies by manufacturer – future work should examine per-device differences [8,50].
χ^2 M1 / M2	44.50*** / 64.00***	7.14 ns / 18.04*	–. Weaker fit reflects sample homogeneity, not model failure. A more demographically diverse Saudi sample would likely yield stronger predictive models [48,52,59].

Table E: Gender Differences Across Key Outcomes

Table 13: Gender Differences Across Key Outcomes

Outcome	Frik et al. [30]	Present Study	Cultural Interpretation
Past behavior (Q9)	Female: ns	No effect	No gender gap in either study. Saudi women’s high smartphone penetration (>90%) masks structural differences in privacy settings knowledge [22,23,60]. Equal intent across genders – privacy protection is a shared value, consistent with Islamic emphasis on Haq al-Khussouiyya as a collective right [16,17,56]. No Saudi gender gap. Saudi women in our sample had similar technical backgrounds to men, possibly reflecting STEM emphasis in Saudi higher education [23]. Consistent across both studies – UX impact perceptions are gender-neutral in both contexts [57]. KEY DIVERGENCE – gender-risk paradox NOT replicated. Saudi women’s concern is grounded in social consequences of location exposure and photo sharing, producing equal concern to men. Challenges universality of the gender-risk paradox across individualist/collectivist cultures [52,53,56,57,60].
Intent (Q12)	Female: ns	No effect	
Difficulty (Q7)	Female: + marginal	No effect	
Negative UX (Q11)	Female: ns	No effect	
Privacy concern (Q15)	Female: LESS concern*** (coef=-0.103)	No effect (coef=-0.21, ns)	

Table F: Context-Aware Design Implications

Table 14: Context-Aware Design Implications

Cultural Dimension	Finding from Present Study	Context-Aware Design Implication
Collectivist Privacy (Khousouiya) [16,17,52,60]	Location & photo privacy valued at household level; Notification preview less configured in shared-device households	Privacy interfaces should offer household-level controls and family-group sharing policies. Context-aware defaults for shared-device environments should be co-designed with Saudi family contexts in mind [31,33].
Linguistic Opacity (Double Translation) [54,55,59]	“Erase data” () ambiguous with factory reset; “Storing audio” () unfamiliar	OS designers must co-design Arabic privacy terminology with native users, replacing transliterations with plain-language equivalents. Cognitive interviews in Arabic should precede any privacy settings UI release [54].
Institutional Trust (Trust Ecosystem) [24,25,58]	High 2FA and ad personalization configuration driven by mandatory government app requirements (Absher, Nafath), not user initiative	Leverage institutional trust — government-endorsed privacy checkups integrated into national digital identity platforms (e.g., Nafath) could drive engagement more effectively than OS-level prompts [24].
Privacy Paradox in Gulf Context [57,60,61]	Storing audio: low configuration despite high awareness; fatalism pattern observed	Design should address self-efficacy barriers directly — progress indicators, confirmation feedback, and reversibility cues for privacy settings would reduce avoidance behavior in Gulf users [33,61].

C Survey Instrument

C.1 Part 1

Q1. What is the operating system on your personal smartphone?

(1 = iOS (only on Apple's iPhones); 2 = Android (e.g., Samsung, Huawei, Google, Xiaomi, etc.); 3 = Other; 4 = I'm not sure)

Q2. In total, how long have you been using smartphones that run this operating system (total time, including phones you previously owned)?

(1 = Less than 1 month; 2 = 2–5 months; 3 = 6–11 months; 4 = 1–2 years; 5 = More than 2 years)

Q3. Introduce any two privacy or security risks you worry about when using your smartphone.

(Consider different actors who may introduce the risks, including companies, governments, someone you know, strangers, etc.)

(open-text: OpenRisk1, OpenRisk2)

Q4. Without checking your phone, describe any settings currently available on your smartphone that you could configure to address the following privacy/security risk: [Risk1/Risk2].

Type "N/A" if you think there are none.

(open-text: OpenSetting1, OpenSetting2)

C.2 Part 2

Well done! In the next part of the survey, you will be asked to answer a series of questions about three smartphone privacy and security settings.

Think about the following smartphone setting: "[Setting]"

Q5. Without looking at your phone, please indicate whether you believe this setting is an existing option on your smartphone.

(1 = Yes; 2 = No; 3 = Not sure)

Q6. If you just bought a phone with this setting as an existing option and had not changed any settings yet, what do you think would be the default configuration?

(1 = Turned on / Allowed; 2 = Turned off / Not allowed; 3 = Not sure; 4 = Not applicable)

Q7. How difficult or easy would it be for you to configure this setting?

(1 = Very easy; 2 = Easy; 3 = Somewhat easy; 4 = Neither difficult nor easy; 5 = Somewhat difficult; 6 = Difficult; 7 = Very difficult)

Q8. Why do you think it would not be easy for you to configure this setting?

(open-text)

Q9. Before taking this survey, have you ever configured this setting on your smartphone?

(1 = Yes; 2 = No; 3 = Not sure)

Q10. Which privacy or security risks would this smartphone setting protect you from?

(open-text)

Q11. If you configure this setting, how would it affect your overall experience of using your smartphone?

(1 = Very negatively; 2 = Negatively; 3 = Somewhat negatively; 4 = Neutral; 5 = Somewhat positively; 6 = Positively; 7 = Very positively; 8 = I don't know)

Q12. How likely would you be to configure this setting?

(1 = Extremely unlikely; 2 = Unlikely; 3 = Somewhat unlikely; 4 = Neither likely nor unlikely; 5 = Somewhat likely; 6 = Likely; 7 = Extremely likely)

Q13. To what extent do you think configuring this smartphone setting would be effective in protecting you from the risks below?

(1 = Very effective; 2 = Effective; 3 = Somewhat effective; 4 = Neither effective nor ineffective; 5 = Somewhat ineffective; 6 = Ineffective; 7 = Very ineffective; 8 = I don't know)

- Data gathering by companies (e.g., for marketing purposes, selling data)
- Data gathering by governments
- Data gathering by individuals (e.g., intimate partner, family member, friend, etc.)
- Unsolicited advertising (e.g., spam)
- Unsolicited objectionable content (e.g., hate speech, fake news, sexually explicit, violent, or otherwise inappropriate content)
- Harassment (e.g., bullying, stalking, blackmailing, doxxing, release of personal information online)
- Social engineering attacks (e.g., scams, phishing)
- Unauthorized financial transactions
- Unauthorized physical access to the device (e.g., theft)
- Unauthorized remote access to the device (e.g., hacking, malware)
- Inadvertent disclosure of personal information

Q14. Do you have any additional comments about this setting?

(open-text)

C.3 Part 3

Q15. How much do you agree or disagree with each of the following statements? “I worry about...”

(1 = Strongly disagree; 2 = Disagree; 3 = Somewhat disagree; 4 = Neutral; 5 = Somewhat agree; 6 = Agree; 7 = Strongly agree; 8 = I’m not familiar with this risk)

- Data gathering by companies
- Data gathering by governments
- Data gathering by individuals
- Unsolicited advertising (spam)
- Objectionable content
- Harassment
- Social engineering attacks
- Unauthorized financial transactions
- Unauthorized physical access
- Unauthorized remote access
- Disclosure of personal information

Q16. Please indicate how easy or difficult it is for you to engage in the following behaviors.

(1 = Very easy; 2 = Easy; 3 = Somewhat easy; 4 = Neutral; 5 = Somewhat difficult; 6 = Difficult; 7 = Very difficult; 8 = I have not tried; 9 = I don’t know)

- To use the Internet to search for information on a smartphone
- To send a text message from a smartphone
- To use a smartphone for entertainment
- To make a video call on a smartphone
- To create a smartphone app
- To schedule an appointment using a health app
- To download an app
- To order food using an app
- To book a ride using an app
- To transfer money using an app

Q17. Do you use a password manager on your smartphone?

(1 = No; 2 = Built-in; 3 = Third-party; 4 = Other; 5 = Not sure)

Q18. What browser do you primarily use on your smartphone?

(1 = Safari; 2 = Chrome; 3 = Firefox; 4 = DuckDuckGo; 5 = Other)

Q19. How often do you use your smartphone?

(1 = Many times a day; 2 = Once or twice a day; 3 = At least once a week; 4 = Less than once a week)

Q20. What is the model name of your smartphone?

(open-text)

Q21. What is the current operating system version of your smartphone?

(open-text)

Q22. Do you use face recognition to unlock your smartphone?

(1 = Yes; 2 = No; 3 = Not sure; 4 = Not available)

C.4 Part 4

Q23. What is your age?

(18–24; 25–34; 35–44; 45–54; 55+)

Q24. What is your gender?

(1 = Male; 2 = Female)

Q25. What language do you primarily use?

(1 = Arabic; 2 = English)

Q26. What language is your smartphone set to?

(1 = Arabic; 2 = English)

Q27. What is your household income?

(Less than 5,000 SAR; 5,000–10,000 SAR; 10,000–15,000 SAR; 15,000–20,000 SAR; 20,000–30,000 SAR; 30,000+ SAR; Prefer not to answer)

Q28. How many people (including you) live in your household?

(One; Two; Three; Four; Five; Six or more)

2785	Q29. Which region do you currently reside in?	2843
2786	(Central; Western; Eastern; Northern; Southern)	2844
2787	Q30. What is the highest level of education you have completed?	2845
2788	(1 = Less than high school; 2 = High school; 3 = Some college; 4 = Associate's; 5 = Bachelor's; 6 = Master's; 7 = Doctoral; 8 = Other; 9 = Prefer	2846
2789	not to answer)	2847
2790	Q31. Do you have a background in any of the following fields?	2848
2791	(Computer science; Software engineering; App development; Other technical; None)	2849
2792	Q32. What is your current employment status?	2850
2793	(Full-time; Part-time; Unemployed looking; Unemployed not looking; Retired; Student; Other)	2851
2794	Q33. Do you have any comments about the survey or study?	2852
2795	(open-text)	2853
2796		2854
2797		2855
2798		2856
2799		2857
2800		2858
2801		2859
2802		2860
2803		2861
2804		2862
2805		2863
2806		2864
2807		2865
2808		2866
2809		2867
2810		2868
2811		2869
2812		2870
2813		2871
2814		2872
2815		2873
2816		2874
2817		2875
2818		2876
2819		2877
2820		2878
2821		2879
2822		2880
2823		2881
2824		2882
2825		2883
2826		2884
2827		2885
2828		2886
2829		2887
2830		2888
2831		2889
2832		2890
2833		2891
2834		2892
2835		2893
2836		2894
2837		2895
2838		2896
2839		2897
2840		2898
2841		2899
2842		2900

D Survey Questions in Arabic

القسم الاول:

1. ما نظام التشغيل في هاتفك الذكي الشخصي؟
(iOS (في هواتف iPhone فقط), Android (مثل Samsung, Huawei, Google, Xiaomi وغيرها), نظام آخر.
(لست متأكدًا)
2. بشكل إجمالي، كم المدة التي استخدمت فيها هواتف تعمل بهذا النظام؟
(بما في ذلك الهواتف السابقة التي امتلكتها) (أقل من شهر، من شهرين إلى 5 أشهر، من 6 إلى 11 شهرًا، من سنة إلى سنتين، أكثر من سنتين)
3. اذكر خطرين متعلقين بالخصوصية والأمان تَقْلُقُ بشأنها عند استخدام هاتفك الذكي. (فكر في الجهات التي قد تسبب هذه المخاطر مثل الشركات، الحكومات، أشخاص تعرفهم، أو أشخاص غريباء)
الخطر الأول: _____
الخطر الثاني: _____
4. دون التحقق من هاتفك، صف أي إعدادات متوفرة حاليًا في هاتفك يمكن استخدامها للتقليل من الخطرين اللذين ذكرتهما.
(إذا كنت تعتقد أنه لا توجد إعدادات، اكتب: لا يوجد)

القسم الثاني: تقييم إعدادات الخصوصية والأمان في الهاتف الذكي

في هذا القسم من الاستبيان سيتم طرح مجموعة من الأسئلة حول ثلاثة إعدادات مختلفة للخصوصية والأمان في الهاتف الذكي.

وصف الإعداد: السماح للتطبيقات بالوصول إلى موقعك الجغرافي (الموقع).

5. دون النظر إلى هاتفك، هل تعتقد أن هذا الإعداد موجود في هاتفك؟

(نعم، لا، لست متأكدًا)

6. إذا اشتريت هاتفًا جديدًا وكان هذا الإعداد موجودًا فيه ولم تقم بتغييره، ماذا تعتقد سيكون الإعداد الافتراضي له؟

(مفتل / مسموح، غير مفتل / غير مسموح، لست متأكدًا، غير منطبق)

7. ما مدى سهولة أو صعوبة ضبط هذا الإعداد بالنسبة لك؟

(1 سهل جدًا، 2، 3، 4، 5، 6، 7 صعب جدًا)

8. لماذا تعتقد أنه قد لا يكون من السهل عليك ضبط هذا الإعداد؟

9. قبل المشاركة في هذا الاستبيان، هل سبق لك ضبط هذا الإعداد في هاتفك؟ (نعم، لا، لست متأكدًا)

10. ما المخاطر المتعلقة بالخصوصية والأمان التي يمكن أن يحميك هذا الإعداد منها؟

11. إذا قمت بتفعيل هذا الإعداد، كيف سيؤثر ذلك على تجربتك في استخدام الهاتف؟ (1 سلبي جدًا، 2، 3، 4، 5، 6، 7 إيجابي جدًا)

12. ما مدى احتمال قيامك بتفعيل هذا الإعداد؟ (1 غير محتمل إطلاقًا، 2، 3، 4، 5، 6، 7 محتمل جدًا)

Figure 8: Survey questions in Arabic: Parts 1 and 2.

13. برأيك، إلى أي مدى يمكن أن يكون هذا الإعداد فعالاً في حمايتك من المخاطر التالية؟ (1 = غير فعال أبداً، 7 = فعال جداً)

a. جمع البيانات من قبل الشركات (مثل التسويق أو بيع البيانات)

b. جمع البيانات من قبل الحكومات

c. جمع البيانات من قبل أفراد (مثل صديق أو صديق أو أحد أفراد العائلة)

d. الإعلانات غير المرغوب فيها (Spam)

e. المحتوى غير اللائق أو المزعج

f. المضايقات أو التتبع أو الابتزاز

g. هجمات الهندسة الاجتماعية (الاحتيال أو التصيد)

h. عمليات مالية غير مصرح بها

i. الوصول غير المصرح به إلى الهاتف ماديًا

j. الوصول غير المصرح به عن بعد (اختراق، فيروسات، برامج خبيثة)

k. الكلفة غير المقصود عن المعلومات الشخصية

14. هل لديك أي تعليقات إضافية حول هذا الإعداد؟

القسم الثالث: المخاوف والقدرات الرقمية

35. ما مدى موافقتك على الجارات التالية: "أنا قلق بشأن..." (1 = لا أوافق بشدة، 7 = أوافق بشدة)

- جمع البيانات من قبل الشركات
- جمع البيانات من قبل الحكومات
- جمع البيانات من قبل أفراد
- الإعلانات المزعجة
- المحتوى غير اللائق
- المضايقات أو التتبع
- هجمات الاحتيال والتصيد
- العمليات المالية غير المصرح بها
- الوصول غير المصرح به إلى الهاتف
- الاختراق والفيروسات
- كلفة المعلومات الشخصية

36. ما مدى سهولة أو صعوبة القيام بالأنشطة التالية باستخدام الهاتف النكي؟ (1 = سهل جداً، 7 = صعب جداً)

- البحث عن معلومات على الإنترنت
- إرسال رسالة نصية
- استخدام الهاتف للترفيه
- إجراء مكالمة فيديو
- إنشاء تطبيق
- حجز موعد طبي عبر تطبيق
- تحميل تطبيق
- طلب طعام عبر تطبيق
- طلب سيارة عبر تطبيق
- تحويل أموال عبر تطبيق

Figure 9: Survey questions in Arabic: Parts 2 (continued) and 3.

القسم الرابع: المعلومات الديموغرافية

37. ما فئتك العمرية؟ (18-24 سنة، 25-34 سنة، 35-44 سنة، 45-54 سنة، 55 سنة فأكثر)
38. ما جنسك؟ (ذكر، أنثى)
39. ما اللغة التي تستخدمها غالباً في حياتك اليومية؟ (العربية، الإنجليزية، أخرى: _____)
40. ما لغة إعدادات هاتفك الذكي؟ (العربية، الإنجليزية، أخرى: _____)
41. ما متوسط الدخل السنوي للأسرة؟ (أقل من 5,000 ريال، من 5,000 إلى 10,000 ريال، من 10,000 إلى 15,000 ريال، من 15,000 إلى 20,000 ريال، من 20,000 إلى 30,000 ريال، 30,000 ريال فأكثر، أفضل عدم الإجابة)
42. كم عدد الأشخاص الذين يعيشون في منزلك (بما فيهم أنت)؟ (تخص واحد، شخصان، 3 أشخاص، 4 أشخاص، 5 أشخاص، 6 أشخاص أو أكثر)
43. ما هي المنطقة التي تقيم فيها حالياً؟ (المنطقة الوسطى (الرياض، القصيم...)، المنطقة الغربية (مكة المكرمة، المدينة المنورة...)، المنطقة الشرقية (الخبر، الجبيل...)، المنطقة الشمالية (تبوك، الحدود الشمالية، الجوف...)، المنطقة الجنوبية (عسير، جازان، نجران، الباحة...))
44. ما أعلى مستوى تعليمي حصلت عليه؟ (أقل من الثانوية، الثانوية، دبلوم، طالب جامعي، بكالوريوس، ماجستير، دكتوراه، أخرى: _____)
45. هل لديك دراسة أو خبرة عمل في المجالات التالية؟ (علوم الحاسب، هندسة البرمجيات، تطوير التطبيقات، مجال تقني آخر، لا شيء مما سبق)
46. ما حالتك الوظيفية الحالية؟ (موظف بدوام كامل، موظف بدوام جزئي، عاطل ويبحث عن عمل، عاطل ولا يبحث عن عمل، متقاعد، طالب، أخرى: _____)
47. هل لديك أي تعليقات حول الاستبيان أو الدراسة؟

Figure 10: Survey questions in Arabic: Part 4.

E Second-Pass Coding of Open-Ended Responses

Table 15: Second-Pass Coding Categories and Their Frequencies

Second Pass Code	Description	Number of Occurrences
Tracking & Surveillance	Participants express concern about being tracked or monitored.	74
Knowledge Gap	Participants are unaware of or unfamiliar with privacy settings.	70
Data Privacy & Leakage	Concern about personal data being exposed or leaked.	34
Hacking & Data Privacy	Concern about hacking leading to data privacy breaches.	34
Cybersecurity Protection	Settings or behaviors used to protect against cyber threats.	32
Voice & Audio Privacy	Concern about audio/voice data being collected without consent.	30
Ad Tracking & Data Privacy	Concern about ad tracking and behavioral profiling.	29
Hacking & Unauthorized Access	Concern about unauthorized access to device or accounts.	26
Data Privacy	General concern about privacy of personal data.	25
App Constraints	Difficulty adjusting settings due to app-specific limitations.	22
Privacy Controls	Using built-in privacy settings to reduce risks.	22
No Knowledge	Participants have no knowledge of relevant settings.	21
Phishing & Malicious Links	Concern about phishing attacks and malicious links.	18
Unauthorized Access	Settings protect against unauthorized access to data or accounts.	18
No Risk	Participants believe a setting provides no real protection.	17
Authentication (2FA)	Using two-factor authentication as a security measure.	17
Navigation & Knowledge Issues	Difficulty finding or navigating to settings.	16
General Security Threat	Vague or general perceived security threat.	15
Vulnerability & Patch Protection	Awareness of software vulnerabilities and patch benefits.	14
Security Awareness	Recognition of need for greater security knowledge.	13
General Security Benefit	Perceived general security benefit of a setting.	12
Account Protection	Settings used to protect user accounts.	12
Vague / No Specific Setting	Participants cannot name a specific setting.	11
Usability & Navigation	Difficulty using or navigating settings due to interface complexity.	10
Tracking & Data Privacy	Combined concern about tracking and data privacy.	9
Hacking & Financial Fraud	Concern about hacking resulting in financial loss.	8
Feature Recommendation	Participants suggest improvements or new features.	7
Confusion & Complexity	Settings are confusing or too complex to adjust.	6
Security Concerns	General unresolved concerns about security.	6
Behavioral Security Practices	Relying on personal behavior rather than settings for security.	5
Device & Account Protection	Settings protecting the physical device and linked accounts.	5
Procedural Complexity	Multi-step processes make settings difficult to configure.	5
Privacy Awareness	Recognition and appreciation of privacy-related settings.	4
Network & Device Risks	Concern about risks from network or device exposure.	4

Second Pass Code	Description	Number of Occurrences
Permission Complexity	Difficulty understanding or managing app permissions.	3
Technical Issues	Technical problems encountered when adjusting settings.	3
Untrusted App Access	Concern about untrusted apps accessing device data.	3
Financial Fraud & Data Privacy	Concern linking financial fraud to data privacy breaches.	3